

Macroscopic system for studies of magnetic dipolar interaction in small particles

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Abstract

A magnetic macroscopic system with free compass needles has been assembled to study dipolar interactions in small particles. Hysteresis loops and the magnetization processes as a function of the system size and dimensions are studied. By changing the number of needles and their disposition, we can vary the magnetostatic coupling and change the magnetic properties of the system. These studies can help to understand better of the role of the magnetostatic coupling in small particle systems.

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The aim of this work is to observe the effects of magnetic dipolar interactions in small particles magnetic systems which continue to be an important branch of studies in magnetism due to its significance either as applied or academic. Small particles magnetic systems can exhibit a different magnetic behavior when compared to bulk materials of the same composition. In small particles systems the exchange interaction is usually higher than the dipolar interaction which is sometimes neglected in computer simulations. Recently, with the appearance of artificially pattern magnetic media, it turned out that dipolar interactions can have a strong effect on the magnetic properties [1]. Guslienko have simulated such array systems considering the magnetostatic energy [2,3]. Ewing [4] and latter Olive and co-authors [5,6] have experimentally studied a magnetic system composed by tiny compass needles arranged in a regular matrix. In this system the dominant magnetic interaction is of dipolar type arising from the magnetostatic coupling between the needles. With this simple system it is possible to measure hysteresis loops and magnetization processes, as well

as to observe the formation of domain like patterns. In our work we have followed the ideas of Olive, and have assembled a macroscopic system with compass needles. We have traced the hysteresis loops and studied its magnetization processes as a function of the system size and geometry. By changing the number of needles and/or its arrangement, we can vary the magnetostatic coupling, which turns out to change the magnetic properties of the system.

The tiny compass needles were acquired commercially and put on non-magnetic sharp pivots. Needles are about 10 mm long and 3 mm wide. Pivots are mount 12 mm apart so that each needle can turn free without touching any of its neighbors. Friction between pivots and needles is unavoidable, but made as small as possible by making the pivot as sharp as possible. An optimized Helmholtz coil produces a homogenous magnetic field at the needles region. The Helmholtz coils are driven by a KEPCO power supply. The magnetization of the system is measured by counting the needles parallel and anti-parallel to the field direction. A 45° angle is considered when the needle points at an intermediate position. In this case, the projection of the magnetization vector in each direction is considered.

Fig. 1 shows the reduced magnetization measured in the field direction. Fig. 1a shows the reduced hysteresis

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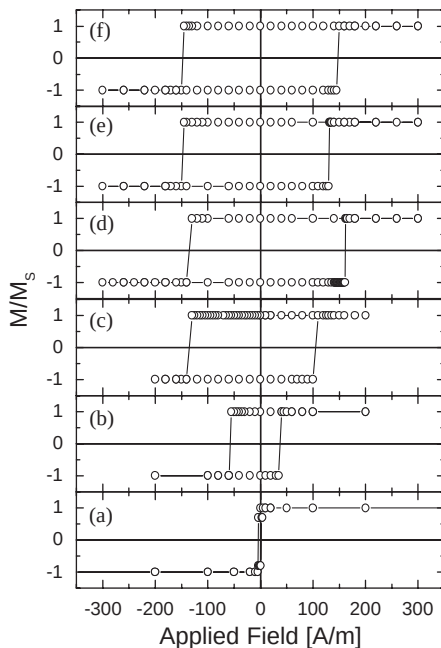


Fig. 1. Reduced magnetization curves measured in aligned needles: (a) one, (b) two, (c) three, (d) four, (e) five and (f) six needles.

loops for one needle alone. The needle is free to rotate about its pivot axis. There is almost no hysteresis. The small irreversibility observed near zero fields is due to the friction between pivot and needle.

In Fig. 1b–f, the number of needles is increased from 2 up to 6, respectively, and the needle arrangement is always linear, as a chain. One can observe here the increase of a coercivity, which is due to the effect of the dipolar type interactions among the needles. Coercivity approaches a saturation value when the number of needles is near 6. Similar results have been observed in other magnetic systems and attributed to dipolar interactions [7]. A 4 needles system can be arranged either in line or in a 2×2 square. Fig. 2 shows the magnetization curves of both. In the in line system the dipolar interaction forces a strong coupling between the needles. In the square system, the dipolar interactions tend to change the coupling as to reduce the coercivity. The direction of the applied field with respect to the needles also affects the magnetic response of the system. Fig. 3 illustrates the effect of a perpendicular field in a 2 and 3 needles systems. Magnetization is measured in the direction of the field. Similar results have also been observed in interacting microwires systems and also explained in terms of dipolar interactions [8].

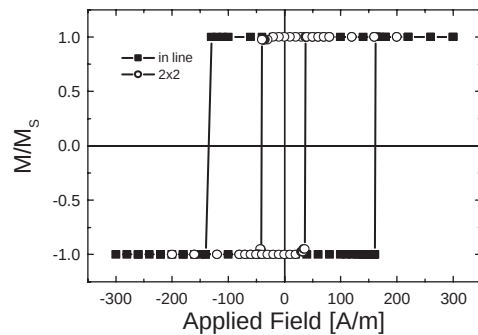


Fig. 2. Magnetization curves for a 4 needles system arranged in line (squares) and in a 2×2 square (open circles).

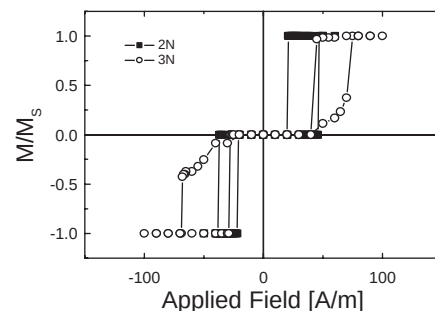


Fig. 3. Reduced magnetization curves obtained by applying the external field perpendicular to the needle line for 2 and 3 needles.

This simple macroscopic system turns out to be a very useful tool to study and observe dipolar type interacting systems. By changes in the geometry and size a quite rich variety of dipolar type phenomena can be observed and studied. This could lead to a better comprehension of the magnetism involved in nanomagnetic materials in which dipolar interactions play a relevant role. This work was supported by CNPq, CAPES (PROCAD NANOMAG), FAPERJ, and Instituto do Milênio de Nanociências.

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